27/8/2026

next class এ আমরা final এ নিয়ে Iterative solution

Inversion Algorithm

A - coefficient matrix

$$A\underline{x} = \underline{b}$$

$$\Rightarrow A^{-1}(A\underline{x}) = A^{-1}(\underline{b})$$

$$\Rightarrow (A^{-1}A)\underline{x} = A^{-1}\underline{b}$$

$$\Rightarrow I\underline{x} = A^{-1}\underline{b}$$

$$\Rightarrow \underline{x} = A^{-1}\underline{b}$$

2. Solve the system by finding A^{-1} by using elementary operations

$$0 + 2x_2 + x_3 + 3x_4 = 1$$
$$x_1 + x_2 - x_3 - 2x_4 = -1$$
$$x_1 + 2x_2 + 0 + x_4 = 2$$
$$-x_1 + x_2 + 2x_3 + 6x_4 = 1$$

Where $A = \begin{pmatrix} 0 & 2 & 1 & 3 \\ 1 & 1 & -1 & -2 \\ 1 & 2 & 0 & 1 \\ -1 & 1 & 2 & 6 \end{pmatrix}$, $\underline{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix}$, $\underline{b} = \begin{pmatrix} 1 \\ -1 \\ 2 \\ 1 \end{pmatrix}$

$$\therefore (A \mid I) = \left(\begin{array}{cccc|cccc} 0 & 2 & 1 & 3 & 1 & 0 & 0 & 0 \\ 1 & 1 & -1 & -2 & 0 & 1 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & 1 & 0 \\ -1 & 1 & 2 & 6 & 0 & 0 & 0 & 1 \end{array}\right)$$

$$\sim \left(\begin{array}{cccc|cccc} 1 & 1 & -1 & -2 & 0 & 1 & 0 & 0 \\ 0 & 2 & 1 & 3 & 1 & 0 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & 1 & 0 \\ -1 & 1 & 2 & 6 & 0 & 0 & 0 & 1 \end{array}\right) \quad [R_1 \leftrightarrow R_2]$$

$$\sim \left(\begin{array}{cccc|cccc} 1 & 1 & -1 & -2 & 0 & 1 & 0 & 0 \\ 0 & 2 & 1 & 3 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 3 & 0 & -1 & 1 & 0 \\ 0 & 2 & 1 & 4 & 0 & 1 & 0 & 1 \end{array}\right) \begin{bmatrix} R_3' = R_3 - R_1 \\ R_4' = R_4 + R_1 \end{bmatrix}$$

$$\sim \left(\begin{array}{cccc|cccc} 1 & 1 & -1 & -2 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 3 & 0 & -1 & 1 & 0 \\ 0 & 2 & 1 & 3 & 0 & 0 & 0 & 0 \\ 0 & 2 & 1 & 4 & 0 & 1 & 0 & 1 \end{array}\right) \begin{bmatrix} R_2 \longleftrightarrow R_3 \end{bmatrix}$$

$$\sim \left(\begin{array}{cccc|cccc} 1 & 1 & -1 & -2 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 3 & 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & -3 & 1 & 2 & -2 & 0 \\ 0 & 0 & -1 & -2 & 0 & 3 & -2 & 1 \end{array}\right) \begin{bmatrix} R_3' = R_3 - 2R_2 \\ R_4' = R_4 - 2R_3 \end{bmatrix}$$

$$\sim \left(\begin{array}{cccc|cccc} 1 & 1 & -1 & -2 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 3 & 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & -3 & 1 & 2 & -2 & 0 \\ 0 & 0 & 0 & 1 & -1 & 1 & 0 & 1 \end{array}\right) \begin{bmatrix} R_4' = R_4 - R_3 \end{bmatrix}$$

$$\sim \left(\begin{array}{cccc|cccc} 1 & 1 & -1 & -2 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 3 & 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & 3 & -1 & -2 & 2 & 0 \\ 0 & 0 & 0 & 1 & -1 & 1 & 0 & 1 \end{array}\right) \begin{bmatrix} R_3' = -R_3 \end{bmatrix}$$

$$\sim \left(\begin{array}{cccc|cccc} 1 & 1 & -1 & 0 & -2 & 3 & 0 & 2 \\ 0 & 1 & 1 & 0 & 3 & -4 & 1 & -3 \\ 0 & 0 & 1 & 0 & 2 & -5 & 2 & -3 \\ 0 & 0 & 0 & 1 & -1 & 1 & 0 & 1 \end{array}\right) \begin{bmatrix} R_1' = R_1 + 2R_4 \\ R_2' = R_2 - 3R_4 \\ R_3' = R_3 - 3R_4 \end{bmatrix}$$

$$\sim \left(\begin{array}{cccc|cccc} 1 & 1 & 0 & 0 & 0 & -2 & 2 & -1 \\ 0 & 1 & 0 & 0 & 1 & 1 & -1 & 0 \\ 0 & 0 & 1 & 0 & 2 & -5 & 2 & -3 \\ 0 & 0 & 0 & 1 & -1 & 1 & 0 & 1 \end{array}\right) \begin{bmatrix} R_1' = R_1 + R_2 \\ R_2' = R_2 - R_3 \end{bmatrix}$$

$$\sim \left(\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & -1 & -3 & 3 & -1 \\ 0 & 1 & 0 & 0 & 1 & 1 & -1 & 0 \\ 0 & 0 & 1 & 0 & 2 & -5 & 2 & -3 \\ 0 & 0 & 0 & 1 & -1 & 1 & 0 & 1 \end{array}\right) \begin{bmatrix} R_1' = R_1 - R_2 \end{bmatrix}$$

$\sim (I \mid A^{-1})$

$$A^{-1} = \begin{pmatrix} -1 & -3 & 3 & -1 \\ 1 & 1 & -1 & 0 \\ 2 & -5 & 2 & -3 \\ -1 & 1 & 0 & 1 \end{pmatrix}$$

$\rightarrow A\underline{x} = \underline{b}$

$\Rightarrow A^{-1}(A\underline{x}) = A^{-1}\underline{b}$

$\Rightarrow \underline{x} = A^{-1}\underline{b}$

$$\Rightarrow \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} -1 & -3 & 3 & -1 \\ 1 & 1 & -1 & 0 \\ 2 & -5 & 2 & -3 \\ -1 & 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \\ 2 \\ 1 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} -1+3+6-1 \\ 1-1-2 \\ 2+5+4-3 \\ -1-1+0+1 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 7 \\ -2 \\ 8 \\ -1 \end{pmatrix}$$

$3x-3y+4z=1$

$2x-3y+4z=2$

$0-y+z=1$

Here, $A=\begin{pmatrix}3 & -3 & 4\\ 2 & -3 & 4\\ 0 & 1 & 1\end{pmatrix}$, $\underline{X}=\begin{pmatrix}x\\ y\\ z\end{pmatrix}$, $\underline{b}=\begin{pmatrix}1\\ 2\\ 1\end{pmatrix}$

$\therefore (A:I)$

$$=\left(\begin{array}{ccc|ccc}3 & -3 & 4 & 1 & 0 & 0\\ 2 & -3 & 4 & 0 & 1 & 0\\ 0 & 1 & 1 & 0 & 0 & 1\end{array}\right)$$

$$\Rightarrow\left(\begin{array}{ccc|ccc}1 & -1 & \frac{4}{3} & \frac{1}{3} & 0 & 0\\ 2 & -3 & 4 & 0 & 1 & 0\\ 0 & 1 & 1 & 0 & 0 & 1\end{array}\right)\left[R_1'=\frac{R_1}{3}\right]$$

$$\Rightarrow\left(\begin{array}{ccc|ccc}1 & -1 & \frac{4}{3} & \frac{1}{3} & 0 & 0\\ 0 & -1 & \frac{4}{3} & -\frac{2}{3} & 1 & 0\\ 0 & 1 & 1 & 0 & 0 & 1\end{array}\right)\left[R_2'=R_2-2R_1\right]$$

$$\Rightarrow\left(\begin{array}{ccc|ccc}1 & -1 & \frac{4}{3} & \frac{1}{3} & 0 & 0\\ 0 & 1 & -\frac{4}{3} & \frac{2}{3} & -1 & 0\\ 0 & 1 & 1 & 0 & 0 & 1\end{array}\right)\left[R_2'=-1\times R_2\right]$$

$$\Rightarrow\left(\begin{array}{ccc|ccc}1 & -1 & \frac{4}{3} & \frac{1}{3} & 0 & 0\\ 0 & 1 & -\frac{4}{3} & \frac{2}{3} & -1 & 0\\ 0 & 0 & \frac{7}{3} & -\frac{2}{3} & 1 & \end{array}\right)\left[R_3'=R_3-R_2\right]$$

$$\Rightarrow\left(\begin{array}{ccc|ccc}1 & -1 & \frac{4}{3} & \frac{1}{3} & 0 & 0\\ 0 & 1 & -\frac{4}{3} & \frac{2}{3} & -1 & 0\\ 0 & 0 & 1 & -\frac{2}{7} & \frac{3}{7} & \frac{3}{7}\end{array}\right)\left[R_3'=R_3\times\frac{3}{7}\right]$$

$$\Rightarrow \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & -1 & 0 \\ 0 & 1 & -\frac{1}{3} & \frac{2}{3} & -1 & 0 \\ 0 & 0 & 1 & -\frac{2}{7} & \frac{3}{7} & \frac{3}{7} \end{array}\right) \quad [R_1' = R_1 + R_2]$$

$$\Rightarrow \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & -1 & 0 \\ 0 & 1 & 0 & \frac{2}{7} & -\frac{3}{7} & \frac{4}{7} \\ 0 & 0 & 1 & -\frac{2}{7} & \frac{3}{7} & \frac{3}{7} \end{array}\right) \quad [R_2' = R_2 + R_3 \times \frac{4}{3}]$$

$\rightarrow A\underline{x} = \underline{b}$

$\Rightarrow A^{-1}(A\underline{x}) = A^{-1}\underline{b}$

$\Rightarrow \underline{x} = A^{-1}b$

$$\therefore A^{-1} = \begin{pmatrix} 1 & -1 & 0 \\ \frac{2}{7} & -\frac{3}{7} & \frac{4}{7} \\ -\frac{2}{7} & \frac{3}{7} & \frac{3}{7} \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 & -1 & 0 \\ \frac{2}{7} & -\frac{3}{7} & \frac{4}{7} \\ -\frac{2}{7} & \frac{3}{7} & \frac{3}{7} \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$$



$$\Rightarrow \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 - 2 + 0 \\ -\frac{2}{7} - \frac{6}{7} + \frac{4}{7} \\ -\frac{2}{7} + \frac{6}{7} + \frac{3}{7} \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$$

H.W

(1) $5x - 7y + z = 11$
$6x - 8y - z = 15$
$3x + 2y - 6z = 7$

(2) $5x - 6y + 4z = 15$
$7x + 4y - 3z = 19$
$2x + y + 6z = 16$

H·W (starts here)

① $5x - 7y + z = 11$
$6x - 8y - z = 15$
$3x + 2y - 6z = 7$

Here,
$A = \begin{bmatrix} 5 & -7 & 1 \\ 6 & -8 & -1 \\ 3 & 2 & -6 \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$, $b = \begin{bmatrix} 11 \\ 15 \\ 7 \end{bmatrix}$

$\therefore [A|I]$

$$\Rightarrow \left[\begin{array}{ccc|ccc} 5 & -7 & 1 & 1 & 0 & 0 \\ 6 & -8 & -1 & 0 & 1 & 0 \\ 3 & 2 & -6 & 0 & 0 & 1 \end{array}\right]$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & -1 & -2 & -1 & 1 & 0 \\ 6 & -8 & -1 & 0 & 1 & 0 \\ 3 & 2 & -6 & 0 & 0 & 1 \end{array}\right] \quad [R_1' = R_2 - R_1]$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & -1 & -2 & -1 & 1 & 0 \\ 0 & -2 & 11 & 6 & -5 & 0 \\ 0 & 5 & 0 & 3 & -3 & 1 \end{array}\right] \quad \begin{bmatrix} R_2' = R_2 - 6R_1 \\ R_3' = R_3 - 3R_1 \end{bmatrix}$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & -1 & -2 & -1 & 1 & 0 \\ 0 & 5 & 0 & 3 & -3 & 1 \\ 0 & -2 & 11 & 6 & -5 & 0 \end{array}\right] \quad [R_2 \leftrightarrow R_3]$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & -1 & -2 & -1 & 1 & 0 \\ 0 & 1 & 0 & \frac{3}{5} & \frac{-3}{5} & \frac{1}{5} \\ 0 & -2 & 11 & 6 & -5 & 0 \end{array}\right] \quad [R_2' = \frac{R_2}{5}]$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & 0 & -2 & -\frac{2}{5} & \frac{2}{5} & \frac{1}{5} \\ 0 & 1 & 0 & \frac{3}{5} & -\frac{3}{5} & \frac{1}{5} \\ 0 & 0 & 11 & 36/5 & -\frac{3}{5} & \frac{2}{5} \end{array}\right] \quad \begin{bmatrix} R_1' = R_1 + R_2 \\ R_3' = R_3 + 2R_2 \end{bmatrix}$$

H.W

$$\sim \left[\begin{array}{ccc|ccc} 1 & 0 & -2 & -2/5 & 2/5 & 1/5 \\ 0 & 1 & 0 & 3/5 & -3/5 & 1/5 \\ 0 & 0 & 1 & 36/55 & -31/55 & 2/55 \end{array}\right] \quad \left[R_3' = \frac{R_3}{11}\right]$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 10/11 & -8/11 & 3/11 \\ 0 & 1 & 0 & 3/5 & -3/5 & 1/5 \\ 0 & 0 & 1 & 36/55 & -31/55 & 2/55 \end{array}\right] \quad \left[R_1' = R_1 + 2R_3\right]$$

$$A^{-1} = \begin{bmatrix} 10/11 & -8/11 & 3/11 \\ 3/5 & -3/5 & 1/5 \\ 36/55 & -31/55 & 2/55 \end{bmatrix}$$

$\rightarrow A\underline{x} = \underline{b}$

$\Rightarrow A^{-1}(A\underline{x}) = A^{-1}\underline{b}$

$\Rightarrow \underline{x} = A^{-1}\underline{b}$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 10/11 & -8/11 & 3/11 \\ 3/5 & -3/5 & 1/5 \\ 36/55 & -31/55 & 2/55 \end{bmatrix} \begin{bmatrix} 11 \\ 15 \\ 7 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ -1 \end{bmatrix}$$

H.W

② $5x-6y+4z=15$
$7x+4y-3z=19$
$2x+y+6z=46$

Here, $A=\begin{bmatrix}5 & -6 & 4\\ 7 & 4 & -3\\ 2 & 1 & 6\end{bmatrix}$, $x=\begin{bmatrix}x\\ y\\ z\end{bmatrix}$, $b=\begin{bmatrix}15\\ 19\\ 46\end{bmatrix}$

$\therefore [A|I]$

$$\Rightarrow \left[\begin{array}{ccc|ccc} 5 & -6 & 4 & 1 & 0 & 0\\ 7 & 4 & -3 & 0 & 1 & 0\\ 2 & 1 & 6 & 0 & 0 & 1\end{array}\right]$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & -8 & -8 & 1 & 0 & -2\\ 7 & 4 & -3 & 0 & 1 & 0\\ 2 & 1 & 6 & 0 & 0 & 1\end{array}\right] \quad [R_1' = R_1 - 2R_3]$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & -8 & -8 & 1 & 0 & -2\\ 0 & 60 & 53 & -7 & 1 & 14\\ 0 & 17 & 22 & -2 & 0 & 5\end{array}\right] \quad \begin{bmatrix}R_2' = R_2 - 7R_1\\ R_3' = R_3 - 2R_1\end{bmatrix}$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & -8 & -8 & 1 & 0 & -2\\ 0 & 1 & 53/60 & -7/60 & 1/60 & 14/60\\ 0 & 17 & 22 & -2 & 0 & 5\end{array}\right] \quad \left[R_2' = \frac{-R_2}{60}\right]$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & 0 & -14/15 & 1/15 & 2/15 & -2/15\\ 0 & 1 & 53/60 & -7/60 & 1/60 & 14/60\\ 0 & 0 & 419/60 & -1/60 & -17/60 & 31/30\end{array}\right] \quad \begin{bmatrix}R_1' = R_1 + 8R_2\\ R_3' = R_3 - 17R_2\end{bmatrix}$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & 0 & -14/15 & 1/15 & 2/15 & -2/15\\ 0 & 1 & 53/60 & -7/60 & 1/60 & 14/60\\ 0 & 0 & 1 & -1/419 & -17/419 & 62/419\end{array}\right] \quad \left[R_3' = \frac{60}{419}R_3\right]$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 27/419 & 40/419 & 2/419\\ 0 & 1 & 0 & -48/419 & 22/419 & 43/419\\ 0 & 0 & 1 & -1/419 & -17/419 & 62/419\end{array}\right] \quad \begin{bmatrix}R_1' = R_1 + \frac{14}{15}R_3\\ R_2' = R_2 - \frac{53}{60}R_3\end{bmatrix}$$

$\sim [I|A^{-1}]$

H.w (Ends here)

$$\therefore A^{-1} = \begin{bmatrix} \frac{27}{419} & \frac{40}{419} & \frac{2}{419} \\ -\frac{48}{419} & \frac{22}{419} & \frac{43}{419} \\ -\frac{1}{419} & \frac{-17}{419} & \frac{62}{419} \end{bmatrix}$$

$\rightarrow A\underline{x} = \underline{b}$

$\Rightarrow A^{-1}(A\underline{x}) = A^{-1}\underline{b}$

$\Rightarrow \underline{x} = A^{-1}\underline{b}$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 27/419 & 40/419 & 2/419 \\ -48/419 & 22/419 & 43/419 \\ -1/419 & -17/419 & 62/419 \end{bmatrix} \begin{bmatrix} 15 \\ 19 \\ 96 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} \frac{405}{419} + \frac{760}{419} + \frac{92}{419} \\ -\frac{720}{419} + \frac{418}{419} + \frac{1978}{419} \\ -\frac{15}{419} - \frac{323}{419} + \frac{2852}{419} \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \\ 6 \end{bmatrix}$$

27/9/2026

Solution of Simultaneous Linear equation by determinants [Cramner's Rule]

Let us solve the following equations:

$$a_1x+b_1y+c_1z=d_1$$
$$a_2x+b_2y+c_2z=d_2$$
$$a_3x+b_3y+c_3z=d_3$$

$$D=\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} \neq 0$$

← 0 হলে Cramner's Rule দিয়ে করা সম্ভব না

$$D_1=\begin{vmatrix} d_1 & b_1 & c_1 \\ d_2 & b_2 & c_2 \\ d_3 & b_3 & c_3 \end{vmatrix}$$

$$D_2=\begin{vmatrix} a_1 & d_1 & c_1 \\ a_2 & d_2 & c_2 \\ a_3 & d_3 & c_3 \end{vmatrix}$$

$$D_3=\begin{vmatrix} a_1 & b_1 & d_1 \\ a_2 & b_2 & d_2 \\ d_3 & b_3 & d_3 \end{vmatrix}$$

$$x=\frac{D_1}{D}$$

$$y=\frac{D_2}{D}$$

$$z=\frac{D_3}{D}$$

$5x - 7y + z = 11$

$6x - 8y - z = 15$

$3x + 2y - 6z = 7$

Soln:-

From the given equations

$$D = \begin{vmatrix} 5 & -7 & 1 \\ 6 & -8 & -1 \\ 3 & 2 & -6 \end{vmatrix} = 55$$

$$D_1 = \begin{vmatrix} 11 & -7 & 1 \\ 15 & -8 & -1 \\ 7 & 2 & -6 \end{vmatrix} = 55$$

$$D_2 = \begin{vmatrix} 5 & 11 & 1 \\ 6 & 15 & -1 \\ 3 & 7 & 6 \end{vmatrix} = -55$$

$$D_3 = \begin{vmatrix} 5 & -7 & 11 \\ 6 & -8 & 15 \\ 3 & 2 & 7 \end{vmatrix} = -55$$

By cramer's Rule,

$$x = \frac{D_1}{D} = \frac{55}{55} = 1$$

$$y = \frac{D_2}{D} = \frac{-55}{55} = -1$$

$$z = \frac{D_3}{D} = \frac{-55}{55} = -1$$

Hence, $x = 1$, $y = -1$, $z = -1$

Ex:- Solve the following system of equations by using determinants

$$x+y+z=1$$
$$ax+by+cz=k$$
$$a^2x+b^2y+c^2z=k^2$$

Sol^n:

We have $D=\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix}$

$$=\begin{vmatrix} 1 & 0 & 0 \\ a & b-a & c-a \\ a^2 & b^2-a^2 & c^2-a^2 \end{vmatrix} \quad \left[\begin{matrix} C_2' = C_2 - C_1 \\ C_3' = C_3 - C_1 \end{matrix}\right]$$

$$=\begin{vmatrix} 1 & 0 & 0 \\ a & b-a & c-a \\ a^2 & (b+a)(b-a) & (c+a)(c-a) \end{vmatrix}$$

$$=(b-a)(c-a)\begin{vmatrix} 1 & 0 & 0 \\ a & 1 & 1 \\ a^2 & b+a & c+a \end{vmatrix}$$

$$=(b-a)(c-a)\cdot 1\begin{vmatrix} 1 & 1 \\ b+a & c+a \end{vmatrix}$$

$$=(b-a)(c-a)(c+a-b-a)$$
$$=(b-a)(c-a)(c-b)$$
$$=(b-c)(c-a)(a-b)$$

Now,

$$D_1 = \begin{vmatrix} 1 & 1 & 1 \\ k & b & c \\ k^2 & b^2 & c^2 \end{vmatrix}$$ ~~[Replacing~~

~~Replacing a by~~

Replacing a by k in (1)

$$D_1 = (b-c)(c-k)(k-b)$$

$$D_2 = \begin{vmatrix} 1 & 1 & 1 \\ a & k & c \\ a^2 & k^2 & c^2 \end{vmatrix}$$

Replacing b by k in (1), $D_2 = (k-c)(c-a)(a-k)$

$$D_3 = \begin{vmatrix} 1 & 1 & 1 \\ a & b & k \\ a^2 & b^2 & k^2 \end{vmatrix}$$

Replacing c by k in (1), $D_3 = (a-b)(b-k)(k-a)$

$$\therefore x = \frac{D_1}{D} = \frac{(b-c)(c-k)(k-b)}{(b-c)(c-a)(a-b)} = \frac{(c-k)(k-b)}{(c-a)(a-b)}$$

$$\therefore y = \frac{D_2}{D} = \frac{(k-c)(c-a)(a-k)}{(b-c)(c-a)(a-b)} = \frac{(k-c)(a-k)}{(b-c)(a-b)}$$

$$\therefore z = \frac{D_3}{D} = \frac{(a-b)(b-k)(k-a)}{(b-c)(c-a)(a-b)} = \frac{(b-k)(k-a)}{(b-c)(c-a)}$$

Determinant থেকে 10 marks এর একটি Determinant থেকে final এ আসবে।

Ex:-
The sum of three numbers is 6. If we multiply the third number by 2 and and add the first number to the result, we get 7. By adding second and third numbers to three times the first number we get 12. Using determinant find the numbers.

Soln:-
Let the three numbers be x, y and z
Then from the given conditions, we have,

$$x+y+z=6$$
$$x+2z=7$$
$$3x+y+z=12$$

Here,

$$D=\begin{vmatrix}1 & 1 & 1\\ 1 & 0 & 2\\ 3 & 1 & 1\end{vmatrix}=4$$

$$D_1=\begin{vmatrix}6 & 1 & 1\\ 7 & 0 & 2\\ 12 & 1 & 1\end{vmatrix}=12$$

$$D_2=\begin{vmatrix}1 & 6 & 1\\ 1 & 7 & 2\\ 3 & 12 & 1\end{vmatrix}=4$$

$$D_3=\begin{vmatrix}1 & 1 & 6\\ 1 & 0 & 7\\ 3 & 1 & 12\end{vmatrix}=8$$

$$\therefore x=\frac{D_1}{D}=\frac{12}{4}=3$$

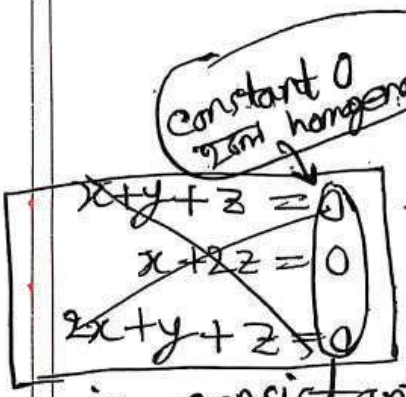
$$\therefore y=\frac{D_2}{D}=\frac{4}{4}=1 \qquad z=\frac{D_3}{D}=\frac{8}{4}=2$$

Thus the three numbers are 3, 1, 2

Homogeneous System

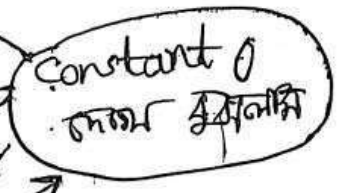
$D = 0$
Consistent with infinitely many solutions.

$D \neq 0$
Consistant with unique solution
$x=0, y=0, z=0$
(Trivial Solution)



Ex:
Find the values of λ, for which the following system of equations is consistant and has non-trivial solution.

$$(\lambda-1)x + (3\lambda+1)y + 2\lambda z = 0$$
$$(\lambda-1)x + (4\lambda-2)y + (\lambda+3)z = 0$$
$$2x + (3\lambda+1) + 3(\lambda-1)z = 0$$



Solution:
This is a system of homogeneous equations.
For non-trivial solution, $D = 0$.

$$\therefore D = \begin{vmatrix} \lambda-1 & 3\lambda+1 & 2\lambda \\ \lambda-1 & 4\lambda-2 & \lambda+3 \\ 2 & 3\lambda+1 & 3\lambda-3 \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} 0 & -\lambda+3 & \lambda-3 \\ \lambda-1 & 4\lambda-2 & \lambda+3 \\ 2 & 3\lambda+1 & 3\lambda-3 \end{vmatrix} = 0 \quad [R_1' = R_1 - R_2]$$

$$\Rightarrow \begin{vmatrix} 0 & 0 & \lambda-3 \\ \lambda-1 & 5\lambda+1 & \lambda+3 \\ 2 & 6\lambda-2 & 3\lambda-3 \end{vmatrix} = 0 \quad [C_2' = C_2 + C_3]$$

$$\Rightarrow (\lambda-3)\left[(\lambda-1)(6\lambda-2) - 2(5\lambda+1)\right] = 0$$

$$\Rightarrow \left[6\lambda^2 - 8\lambda + 2 - 10\lambda - 2\right] = 0$$

$$\Rightarrow 6\lambda^2 - 18\lambda = 0$$

$$\Rightarrow 6\lambda(\lambda-3) = 0$$

$$\therefore \lambda = 3, 0$$

29/9/2026

A system of linear equation has either

① no solution
② Exactly one solution
③ infinitely many solutions

⊛ A system of linear equations is called inconsistent if it has no solution

⊛ A system of linear equations is said to be consistent if it has either one solution or infinitely many solution

Gaussian Elimination method

Consider

$$a_1x_1 + b_1x_2 + c_1x_3 = d_1$$
$$a_2x_1 + b_2x_2 + c_2x_3 = d_2$$
$$a_3x_1 + b_3x_2 + c_3x_3 = d_3$$

⊛ Step 1: Form the argmented matrix

$$\left(\begin{array}{ccc|c} a_1 & b_1 & c_1 & d_1 \\ a_2 & b_2 & c_2 & d_2 \\ a_3 & b_3 & c_3 & d_3 \end{array}\right)$$

Step 2: Form the row echelon form by elementary row operation

$$\left(\begin{array}{ccc|c} a_1 & b_1 & c_1 & d_1 \\ 0 & b_2' & c_2' & d_2' \\ 0 & 0 & c_3' & d_3' \end{array}\right)$$

Step 3:-

In equation form,

$$a_1 x_1 + b_1 x_2 + c_1 x_3 = d_1$$
$$b_2' x_2 + c_2' x_3 = d_2'$$
$$c_3' x_3 = d_3'$$

Which is the $A\underline{x} = \underline{b}$ form

where, $A = \begin{pmatrix} a_1 & b_1 & c_1 \\ 0 & b_2' & c_2' \\ 0 & 0 & c_3' \end{pmatrix}$, $\underline{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$; $\underline{b} = \begin{pmatrix} d_1 \\ d_2 \\ d_3 \end{pmatrix}$

Step 4:-

By back substitution we get the result

① $x+y+z=1$

$3x-y-z=4$

$x+5y+5z=-1$

Soln:-

The augmented matrix is

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 3 & -1 & -1 & 4 \\ 1 & 5 & 5 & -1 \end{array}\right)$$

By elementary row operation,

$$\sim \left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & -4 & -4 & 1 \\ 0 & 4 & 4 & -2 \end{array}\right) \begin{bmatrix} R_2' = R_2 - 3R_1 \\ R_3' = R_3 - R_1 \end{bmatrix}$$

$$\sim \left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & -4 & -4 & 1 \\ 0 & 0 & 0 & -1 \end{array}\right) \quad [R_3' = R_3 + R_2]$$

In matrix form,

$$x+y+z=1$$
$$-4y-4z=1$$
$$0=-1$$

$0=-1$ is not true

The system is inconsistent

Thus there is no solution for this system.

অগ্রাধিকার না, যেমন এভাবেও করা যায়

② $2x+y-2z=10$
$3x+2y+2z=1$
$5x+4y+3z=4$

Soln:-
Reduce the system to echelon form by means of elementary operation,

$$\sim \begin{cases} 2x+y-2z=10 \\ y+10z=-28 \\ 3y+16z=-42 \end{cases} \quad \begin{bmatrix} L_2' = 2L_2 - 3L_1 \\ L_3' = 2L_3 - 5L_1 \end{bmatrix}$$

↖ L means line

$$\sim \begin{cases} 2x+y-2z=10 \\ y+10z=-28 \\ -14z=42 \end{cases} \quad [L_3' = L_3 - 3L_2]$$

By back substitution we get

$-14z = 42$
$\therefore z = -3$

$y = -10z - 28 = -10(-3) - 28 = 2$

$2x = -y + 2z + 10 = -2 - 6 + 10 = -12 + 10 = -2$
$\therefore x = -1$

Ans: $(-1, 2, -3)$

System solution এর উদাহরণ CT3 2016।

~~system এবারের প্রশ্ন~~

③ $x+2y-3z=-2$
$3x-y-2z=1$
$2x+3y-5z=-3$

Soln:-

The augmented matrix is

$$\left(\begin{array}{ccc|c} 1 & 2 & -3 & -2 \\ 3 & -1 & -2 & 1 \\ 2 & 3 & -5 & -3 \end{array}\right)$$

$$\sim \left(\begin{array}{ccc|c} 1 & 2 & -3 & -2 \\ 0 & -7 & 7 & 7 \\ 0 & -1 & 1 & 1 \end{array}\right) \begin{bmatrix} R_2' = R_2 - 3R_1 \\ R_3' = R_3 - 2R_1 \end{bmatrix}$$

$$\sim \left(\begin{array}{ccc|c} 1 & 2 & -3 & -2 \\ 0 & 1 & -1 & -1 \\ 0 & -1 & 1 & 1 \end{array}\right) \left[R_2' = -\frac{1}{7} R_2\right]$$

$$\sim \left(\begin{array}{ccc|c} 1 & 2 & -3 & -2 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & 0 & 0 \end{array}\right) \left[R_3' = R_2 + R_3\right]$$

The corresponding system is

$$x+2y-2z=-2$$
$$y-z=-1$$

Thus system has two equations and three variable. The free variable $= (3-2)=1$.
Let the free variable be z
Let $z=5$
$\therefore y=4, x=5$

~~particular solution / Infinitely many solution~~
particular solution $(5,4,5)$

parametric solution

Let $z = t$,

$\therefore y = -1 + t$

$\therefore x = -2 - 2y + 3z$

$= -2 - 2(-1+t) + 3t$

$= -2 + t - 2t + 3t$

$= t$

$\therefore (t, (-1+t), t)$

C.W

① $2x - 3y + 5z = 1$

$3x + y - z = 2$

$x + 4y - 6z = 1$

② $x - 4y + 5z = 8$

$3x + 7y - z = 3$

$x + 15y - 11z = -14$

③ $x + 2y + 3z = 0$

$2x + 5y + 2z = 0$

$3x - y - 4z = 0$

H.W.

~~Exercises~~

① $2x-3y+5z=1$

$3x+y-z=2$

$x+4y-6z=1$

We have, $D=\begin{vmatrix} 2 & -3 & 5 \\ 3 & 1 & -1 \\ 1 & 4 & -6 \end{vmatrix}=0$ (সমাধান হবে না)

Reduce the system to echelon form by means of elementary operation

$$\sim \begin{cases} 2x-3y+5z=1 \\ 11y-17z=1 \\ 11y-17z=1 \end{cases} \quad \begin{bmatrix} L_2'=2L_2-3L_1 \\ L_3'=2L_3-L_1 \end{bmatrix}$$

$$\sim \begin{cases} 2x-3y+5z=1 \\ 11y-17z=1 \\ 0=0 \end{cases} \quad [L_3'=L_3-L_2]$$

The corresponding system is

$$2x-3y+5z=1$$
$$11y-17z=1$$

Thus the system has two equations and three variable

The free variable $=(3-2)=1$

Let the free variable be z

Let $z=4$

$\therefore y=6.2727272727272727$, $z=-0.0909090909$

particular solution $(-0.09, 6.27, 4)$

Parametric solution

Let $z=t$

$\therefore y=\frac{1+17t}{11}$

$2x=3y-5z+1=3\left(\frac{1+17t}{11}\right)-5z+1$

$\Rightarrow x=\frac{3+51t-55z+11}{22}$

$=\frac{14-4t}{22}=\frac{7-2t}{11}$

$\therefore \left(\frac{14-4t}{22}, \frac{1+17t}{11}, t\right)$

H.W

(2) $x-4y+5z=8$

$3x+7y-z=3$

$x+15y-11z=-19$

Reduce the system to echelon form by means of elementary operation

$$\sim \begin{cases} x-4y+5z=8 \\ 19y-16z=-21 \\ 19y-16z=-22 \end{cases} \quad \begin{bmatrix} L_2' = L_2 - 9L_1 \\ L_3' = L_3 - L_1 \end{bmatrix}$$

$$\sim \begin{cases} x-4y+5z=8 \\ 19y-16z=-21 \\ 0=-1 \end{cases} \quad \begin{bmatrix} L_3' = L_3 - L_2 \end{bmatrix}$$

$0=-1$ is not true

the system is inconsistent

Thus there is no solution for this system

H.W

③ $x+2y+3z=0$
$2x+5y+2z=0$
$3x-y-4z=0$

Reduce the system to echelon form by means of elementary operations

$$\sim \begin{cases} x+2y+3z=0 \\ y-4z=0 \\ -7y-13z=0 \end{cases} \quad \begin{bmatrix} R_2'=R_2-2R_1 \\ R_3'=R_3-3R_1 \end{bmatrix}$$

$$\sim \begin{cases} x+2y+3z=0 \\ y-4z=0 \\ -41z=0 \end{cases} \quad \left[R_3'=R_3+7R_2\right]$$

By back substitution we get

$-41z=0$

$\Rightarrow z=0$

$y-4\times 0=0$

$\Rightarrow y=0$

$x+2\times 0+3\times 0=0$

$\Rightarrow x=0$

Ans $(0,0,0)$

A1 10/26

Solve the following system

(1) $2x+4y+6z=22$
$3x+8y+5z=27$
$-x+y+2z=2$

Ans: $(3, 1, 2)$

(2) $3x-2y+8z=9$
$-2x+2y+z=3$
$x+2y-3z=8$

Ans $(3, 4, 1)$

(3) Find the parametric solution of

$x+2y-3z=-2$
$3x-y-2z=1$
$2x+3y-5z=-3$

Ans: $\{(t, t-1, t): t \in \mathbb{R}$

(4) $x-4y+5z=8$
$3x+7y-z=3$
$x+15y-11z=-14$

No solution

(5) Find the particular solution of

$x_1+2x_2-3x_3=6$
$2x_1-x_2+4x_3=2$
$4x_1+3x_2-2x_3=14$

① $2x+4y+6z=22$
$3x+8y+5z=27$
$-x+y+2z=2$

Reduce the system to echelon form by means of elementary operations

$$\sim \begin{cases} 2x+4y+6z=22 \\ 4y-8z=-12 \\ 6y+10z=26 \end{cases} \quad [R_2' = 2R_2 - 3R_1,\ R_3' = 2R_3 + R_1]$$

$$\sim \begin{cases} 2x+4y+6z=22 \\ 4y-8z=-12 \\ 44z=88 \end{cases} \quad [R_3' = 2R_3 - 3R_2]$$

By back substitution we get

$44z=88$

$\Rightarrow z=\frac{88}{44}=2$

$4y-8\times 2=-12$

$\Rightarrow 4y=-12+16$

$\Rightarrow y=1$

$2x+4\times 1+6\times 2=22$

$\Rightarrow 2x+16=22$

$\Rightarrow 2x=6$

$\Rightarrow x=3$

$\therefore$ Ans $=(3,1,2)$

② $3x-2y+8z=9$

$-2x+2y+z=3$

$x+2y-3z=8$

Reduce the system to echelon form by means of elementary operations

$$\sim \begin{cases} 3x-2y+8z=9 \\ 2y+19z=27 \\ 8y-17z=15 \end{cases} \quad \begin{bmatrix} R_2' = 3R_2+2R_1 \\ R_3' = 3R_3-R_1 \end{bmatrix}$$

$$\sim \begin{cases} 3x-2y+8z=9 \\ 2y+19z=27 \\ -93z=-93 \end{cases} \quad [R_3' = R_3 - 4R_2]$$

$-93z$

By back substitution we get,

$-93z=-93$

$\Rightarrow z=\frac{-93}{-93}=1$

$2y+19\times1=27$

$\Rightarrow 2y=27-19=8$

$\Rightarrow y=4$

$3x-2\times4+8\times1=9$

$\Rightarrow 3x=9+8-8$

$\Rightarrow x=3$

$\therefore$ Ans $=(3,4,1)$

③ $x+2y-3z=-2$

$3x-y-2z=1$

$2x+3y-5z=-3$

Reduce the system to echelon form by means of elementary operations

$$\sim \begin{cases} x+2y-3z=-2 \\ -7y+7z=7 \\ -y+z=1 \end{cases} \quad \left[\begin{array}{l} R_2' = R_2 - 3R_1 \\ R_3' = R_3 - 2R_1 \end{array}\right]$$

$$\sim \begin{cases} x+2y-3z=-2 \\ -7y+7z=7 \\ 0=0 \end{cases} \quad \left[R_3' = 7R_3 - R_2\right]$$

The corresponding system is

$$x+2y-3z=-2$$
$$-7y+7z=7$$

Thus the system has two equation and three variable.

The free variable $=(3-2)=1$

Let $z=t$

$\therefore -7y+7t=7$

$\Rightarrow -y+t=1$

$\Rightarrow y=t-1$

$\therefore x+2(t-1)-3t=-2$

$\Rightarrow x+2t-2-3t=-2$

$\Rightarrow x-t=0$

$\Rightarrow x=t$

$\therefore$ Ans $(t, t-1, t)$

④ $x-4y+5z=8$

$3x+7y-z=3$

$x+15y-11z=-14$

Reduce the system to echelon form by means of elementary operations

$$\sim \begin{cases} x-4y+5z=8 \\ 19y-16z=-21 \\ 19y-16z=-22 \end{cases} \quad \begin{bmatrix} R_2'=R_2-3R_1 \\ R_3'=R_3-R_1 \end{bmatrix} \quad \begin{bmatrix} L_2'=L_2-3L_1 \\ L_3'=L_3-L_1 \end{bmatrix}$$

$$\sim \begin{cases} x-4y+5z=8 \\ 19y-16z=-21 \\ 0=-1 \end{cases} \quad \begin{bmatrix} R_3'=R_3-R_2 \\ L_3'=L_3-L_2 \end{bmatrix}$$

$0=-1$ is not true

∴ The system is inconsistent

Thus there is no solution for this

⑤ $x_1 + 2x_2 - 3x_3 = 6$

$2x_1 - x_2 + 4x_3 = 2$

$4x_1 + 3x_2 - 2x_3 = 14$

$$\sim \begin{cases} x_1 + 2x_2 - 3x_3 = 6 \\ -5x_2 + 10x_3 = -10 \\ -5x_2 + 10x_3 = -10 \end{cases} \left[\begin{matrix} L_2' = L_2 - 2L_1 \\ L_3' = L_3 - 4L_1 \end{matrix} \right]$$

$$\sim \begin{cases} x_1 + 2x_2 - 3x_3 = 6 \\ -5x_2 + 10x_3 = -10 \\ 0 = 0 \end{cases} \left[L_3' = L_3 - L_2 \right]$$

The corresponding system is

$$x_1 + 2x_2 - 3x_3 = 6$$
$$-5x_2 + 10x_3 = -10$$

Thus system has two equations and three variable. The free variable $= (3-2) = 1$

Let $x_3 = 3$

$\therefore -5x_2 + 10 \times 3 = -10$

$\Rightarrow -5x_2 = -10 - 30$

$\therefore x_2 = 8$

$x_1 + 2 \times 8 - 3 \times 3 = 6$

$x_1 + 16 - 9 = 6$

$\Rightarrow x_1 = -1$

$\therefore$ Particular solution $(-1, 8, 3)$

6/10/2026 পরীক্ষা final ও assignment CT তে না

We call $\underline{b} - A\underline{x}$ the least squares error vector
we call $\|\underline{b} - A\underline{x}\|$ the least squares error

(*) Find the least squares solution/ the least squares error vector/ the least squares error of the linear system.

$$x_1 - x_2 = 4$$
$$3x_1 + 2x_2 = 1$$
$$-2x_1 + 4x_2 = 3$$

Solution:

It will be convenient to express the system in the matrix form $A\underline{x} = \underline{b}$, where

$$A = \begin{pmatrix} 1 & -1 \\ 3 & 2 \\ -2 & 4 \end{pmatrix} \text{ and } \underline{b} = \begin{pmatrix} 4 \\ 1 \\ 3 \end{pmatrix}$$

It follows that

$$A^TA = \begin{pmatrix} 1 & 3 & -2 \\ -1 & 2 & 4 \end{pmatrix}\begin{pmatrix} 1 & -1 \\ 3 & 2 \\ -2 & 4 \end{pmatrix} = \begin{pmatrix} 14 & 3 \\ -3 & 21 \end{pmatrix}$$

$$A^T\underline{b} = \begin{pmatrix} 1 & 3 & -2 \\ -1 & 2 & 4 \end{pmatrix}\begin{pmatrix} 4 \\ 1 \\ 3 \end{pmatrix} = \begin{pmatrix} 1 \\ 10 \end{pmatrix}$$

So the normal system $A^TA\underline{x} = A^T\underline{b}$ is

$$\begin{pmatrix} 14 & -3 \\ -3 & 21 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 10 \end{pmatrix}$$

Solving this system yields an unique least squares solution, namely,

$$x_1 = \frac{17}{95}, \quad x_2 = \frac{143}{285}$$

The least squares error vector is

$$\underline{b} - A\underline{x}$$

$$= \begin{pmatrix} 4 \\ 1 \\ 3 \end{pmatrix} - \begin{pmatrix} 1 & -1 \\ 3 & 2 \\ -2 & 4 \end{pmatrix} \begin{pmatrix} \frac{14}{95} \\ \frac{143}{285} \end{pmatrix}$$

$$= \begin{pmatrix} 4 \\ 1 \\ 3 \end{pmatrix} - \begin{pmatrix} -\frac{92}{285} \\ \frac{439}{285} \\ \frac{95}{57} \end{pmatrix}$$

$$= \begin{pmatrix} \frac{1232}{285} \\ -\frac{154}{285} \\ \frac{4}{3} \end{pmatrix}$$

and the least squares error is

$$\|b - A\underline{x}\| = \sqrt{\left(\frac{1232}{285}\right)^2 + \left(-\frac{154}{285}\right)^2 + \left(\frac{4}{3}\right)^2}$$

$$= 4.556$$

(*) Find the least square solution of $A\underline{x} = \underline{b}$

(1) $A = \begin{pmatrix} 1 & -1 \\ 2 & 3 \\ 4 & 5 \end{pmatrix}$, $\underline{b} = \begin{pmatrix} 2 \\ -1 \\ 5 \end{pmatrix}$

(2) $A = \begin{pmatrix} 2 & -2 \\ 1 & 1 \\ 3 & 1 \end{pmatrix}$, $\underline{b} = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}$

(3) $A = \begin{pmatrix} 1 & 0 & -1 \\ 2 & 1 & -2 \\ 1 & 1 & 0 \\ 1 & 1 & -1 \end{pmatrix}$, $\underline{b} = \begin{pmatrix} 6 \\ 0 \\ 9 \\ 3 \end{pmatrix}$

(4) $A = \begin{pmatrix} 2 & 0 & -1 \\ 1 & -2 & 2 \\ 2 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix}$, $\underline{b} = \begin{pmatrix} 0 \\ 6 \\ 0 \\ 6 \end{pmatrix}$

(1) $A = \begin{pmatrix} 1 & -1 \\ 2 & 3 \\ 4 & 5 \end{pmatrix}$, $\underline{b} = \begin{pmatrix} 2 \\ -1 \\ 5 \end{pmatrix}$

It follows that

$$A^TA = \begin{pmatrix} 1 & 2 & 4 \\ -1 & 3 & 5 \end{pmatrix}\begin{pmatrix} 1 & -1 \\ 2 & 3 \\ 4 & 5 \end{pmatrix} = \begin{pmatrix} 21 & 25 \\ 25 & 35 \end{pmatrix}$$

$$A^T\underline{b} = \begin{pmatrix} 1 & 2 & 4 \\ -1 & 3 & 5 \end{pmatrix}\begin{pmatrix} 2 \\ -1 \\ 5 \end{pmatrix} = \begin{pmatrix} 20 \\ 20 \end{pmatrix}$$

So the normal system ~~yields an unique least~~

$A^TA\underline{x} = A^T\underline{b}$, is

$$\begin{pmatrix} 21 & 25 \\ 25 & 35 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 20 \\ 20 \end{pmatrix}$$

solving the system yields an unique least squares solution namely,

$$x_1 = \frac{20}{11}, \quad x_2 = \frac{-8}{11}$$

The least squares error vector is

$$\underline{b} - A\underline{x}$$

$$= \begin{pmatrix} 2 \\ -1 \\ 5 \end{pmatrix} - \begin{pmatrix} 1 & -1 \\ 2 & 3 \\ 4 & 5 \end{pmatrix}\begin{pmatrix} \frac{20}{11} \\ -\frac{8}{11} \end{pmatrix}$$

$$= \begin{pmatrix} -\frac{6}{11} \\ -\frac{27}{11} \\ \frac{15}{11} \end{pmatrix}$$

and the least squares error is

$$\|\underline{b} - A\underline{x}\| = \sqrt{\left(\frac{-6}{11}\right)^2 + \left(-\frac{27}{11}\right)^2 + \left(\frac{15}{11}\right)^2}$$

$$= \frac{3\sqrt{110}}{11} = 2.86$$

② $A = \begin{pmatrix} 2 & -2 \\ 1 & 1 \\ 3 & 1 \end{pmatrix}$, $\underline{b} = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}$

It follows that

$A^TA = \begin{pmatrix} 2 & 1 & 3 \\ -2 & 1 & 1 \end{pmatrix} \begin{pmatrix} 2 & -2 \\ 1 & 1 \\ 3 & 1 \end{pmatrix} = \begin{pmatrix} 14 & 0 \\ 0 & 6 \end{pmatrix}$

$A^T\underline{b} = \begin{pmatrix} 2 & 1 & 3 \\ -2 & 1 & 1 \end{pmatrix} \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} = \begin{pmatrix} 6 \\ -4 \end{pmatrix}$

So the normal system $A^TA\underline{x} = A^T\underline{b}$ is

$$\begin{pmatrix} 14 & 0 \\ 0 & 6 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 6 \\ -4 \end{pmatrix},$$

Solving the system yield an unique least squares solution namely

$$\therefore x_1 = \frac{3}{7}, \quad x_2 = -\frac{2}{3}$$

The least squares error vector is

$$\underline{b} - A\underline{x} = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} - \begin{pmatrix} 2 & -2 \\ 1 & 1 \\ 3 & 1 \end{pmatrix} \begin{pmatrix} \frac{3}{7} \\ -\frac{2}{3} \end{pmatrix} = \begin{pmatrix} -\frac{4}{21} \\ -\frac{16}{21} \\ \frac{8}{21} \end{pmatrix}$$

And the least squares error is

$$\|\underline{b} - A\underline{x}\| = \sqrt{\left(-\frac{4}{21}\right)^2 + \left(\frac{-16}{21}\right)^2 + \left(\frac{8}{21}\right)^2}$$

$$= \sqrt{\frac{16+256+64}{441}}$$

$$= \sqrt{\frac{336}{441}}$$

$$= \frac{4\sqrt{21}}{21} = 0.87$$

Work at home

③ $A = \begin{pmatrix} 1 & 0 & -1 \\ 2 & 1 & -2 \\ 1 & 1 & 0 \\ 1 & 1 & -1 \end{pmatrix}$, $\underline{b} = \begin{pmatrix} 6 \\ 0 \\ 9 \\ 3 \end{pmatrix}$

It follows that

$$A^TA = \begin{pmatrix} 1 & 2 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ -1 & -2 & 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 & -1 \\ 2 & 1 & -2 \\ 1 & 1 & 0 \\ 1 & 1 & -1 \end{pmatrix}$$

$$= \begin{pmatrix} 1+4+1+1 & 0+2+1+1 & -1-4+0-1 \\ 0+2+1+1 & 0+1+1+1 & 0-2+0-1 \\ -1-4+0-1 & 0-2+0-1 & 1+4+0+1 \end{pmatrix}$$

$$= \begin{pmatrix} 7 & 4 & -6 \\ 4 & 3 & -3 \\ -6 & -3 & 6 \end{pmatrix}$$

$$A^T\underline{b} = \begin{pmatrix} 1 & 2 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ -1 & -2 & 0 & -1 \end{pmatrix} \begin{pmatrix} 6 \\ 0 \\ 9 \\ 3 \end{pmatrix} = \begin{pmatrix} 6+0+9+3 \\ 0+0+9+3 \\ -6-0+0-3 \end{pmatrix} = \begin{pmatrix} 18 \\ 12 \\ -9 \end{pmatrix}$$

So the normal system $A^TA\underline{x} = A^Tb$ is

$$\begin{pmatrix} 7 & 4 & -6 \\ 4 & 3 & -3 \\ -6 & -3 & 6 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 18 \\ 12 \\ -9 \end{pmatrix}$$

$$\Rightarrow \begin{cases} 7x_1 + 4x_2 - 6x_3 = 18 \\ 4x_1 + 3x_2 - 3x_3 = 12 \\ -6x_1 - 3x_2 + 6x_3 = -9 \end{cases}$$

Solving the system yield an unique least squares solution namely

$$x_1 = 12,\ x_2 = -3,\ x_3 = 9$$

The least squares error vector is

$$\underline{b} - A\underline{x} = \begin{pmatrix} 6 \\ 0 \\ 9 \\ 3 \end{pmatrix} - \begin{pmatrix} 1 & 0 & -1 \\ 2 & 1 & -2 \\ 1 & 1 & 0 \\ 1 & 1 & -1 \end{pmatrix} \begin{pmatrix} 12 \\ -3 \\ 9 \end{pmatrix}.$$

$$= \begin{pmatrix} 6 \\ 0 \\ 9 \\ 3 \end{pmatrix} - \begin{pmatrix} 12+0-9 \\ 24-3-18 \\ 12-3+0 \\ 12-3-9 \end{pmatrix}$$

$$\begin{pmatrix} 6 \\ 0 \\ 9 \\ 3 \end{pmatrix} - \begin{pmatrix} 3 \\ 3 \\ 9 \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} 3 \\ -3 \\ 0 \\ 3 \end{pmatrix}$$

and the least squares error is

$$\|\underline{b} - A\underline{x}\| = \sqrt{3^2+(-3)^2+0^2+3^2}$$

$$= \sqrt{9+9+0+9}$$

$$= \sqrt{27}$$

$$= 5.2$$

(4) Given,

$$A = \begin{pmatrix} 2 & 0 & -1 \\ 1 & -2 & 2 \\ 2 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix}, \quad \underline{b} = \begin{pmatrix} 0 \\ 6 \\ 0 \\ 0 \end{pmatrix}$$

It follows that

$$A^TA = \begin{pmatrix} 2 & 1 & 2 & 0 \\ 0 & -2 & -1 & 1 \\ -1 & 2 & 0 & -1 \end{pmatrix} \begin{pmatrix} 2 & 0 & -1 \\ 1 & -2 & 2 \\ 2 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix}$$

$$= \begin{pmatrix} 4+1+4+0 & 0-2-2+0 & -2+2+0+0 \\ 0-2-2+0 & 0+4+1+1 & +0-1-0-1 \\ -2+2+0-0 & -0-1-0-1 & 1+1+0+1 \end{pmatrix}$$

$$= \begin{pmatrix} 9 & -4 & 0 \\ -4 & 6 & -5 \\ 0 & -5 & 6 \end{pmatrix}$$

$$A^T\underline{b} = \begin{pmatrix} 2 & 1 & 2 & 0 \\ 0 & -2 & -1 & 1 \\ -1 & 2 & 0 & -1 \end{pmatrix} \begin{pmatrix} 0 \\ 6 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0+6+0+0 \\ 0-12+0+0 \\ 0+12+0+0 \end{pmatrix} = \begin{pmatrix} 6 \\ -12 \\ 12 \end{pmatrix}$$

So the normal system $A^TA\underline{x} = A^T\underline{b}$ is

$$\begin{pmatrix} 9 & -4 & 0 \\ -4 & 6 & -5 \\ 0 & -5 & 6 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 6 \\ -12 \\ 12 \end{pmatrix}$$

$$= \begin{cases} 9x_1 - 4x_2 = 6 \\ -4x_1 + 6x_2 - 5x_3 = -12 \\ -5x_2 + 6x_3 = 12 \end{cases}$$

Solving the system yield an unique least squares solution namely

$$x_1 = 6, \quad x_2 = 12, \quad x_3 = 12$$

The least error vector is

$$\underline{b} - A\underline{x} = \begin{pmatrix} 0 \\ 6 \\ 0 \\ 0 \end{pmatrix} - \begin{pmatrix} 2 & 0 & -1 \\ 1 & -2 & 2 \\ 3 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix} \begin{pmatrix} 6 \\ 12 \\ 12 \end{pmatrix}$$

$$= \begin{pmatrix} 0 \\ 6 \\ 0 \\ 0 \end{pmatrix} - \begin{pmatrix} 12+0-12 \\ 6-24+24 \\ 12-12+0 \\ 0+12-12 \end{pmatrix}$$

$$= \begin{pmatrix} 0 \\ 6 \\ 0 \\ 0 \end{pmatrix} - \begin{pmatrix} 0 \\ 6 \\ 0 \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

and the least squares error is

$$\|b - A\underline{x}\| = \sqrt{0^2+0^2+0^2+0^2} = 0$$

$5x - 7y + z = 11$

$6x - 8y - z = 15$

$3x + 2y - 6z = 7$

Reduce the system to echelon form by means of elementary operations -

$$\sim \begin{cases} 5x - 7y + z = 11 \\ 2y - 11z = 9 \\ 31y - 33z = 2 \end{cases} \quad \begin{bmatrix} L_2' = 5L_2 - 6L_1 \\ L_3' = 5L_3 - 3L_1 \end{bmatrix}$$

$$\sim \begin{cases} 5x - 7y + z = 11 \\ 2y - 11z = 9 \end{cases} \quad [L_3' = 2L_3 - 31L_2]$$

$$275z = -275$$

By back substitution we get,

$$275z = -275$$

$$\Rightarrow z = \frac{-275}{275} = -1$$

$$2y - 11z = 9$$

$$2 \times y - 11 \times (-1) = 9$$

$$\Rightarrow y = \frac{9 - 11}{2} = \frac{-2}{2} = -1$$

$$5x - 7y + z = 11$$

$$\Rightarrow 5x - 7(-1) + (-1) = 11$$

$$\Rightarrow 5x + 7 - 1 = 11$$

$$\Rightarrow 5x = 11 - 7 + 1 = 5$$

$$\Rightarrow x = 1$$

$\therefore$ Ans $(1, -1, -1)$

13/10/2026

এই বার্ষি সাপ্তাহে final এ ~~[illegible]~~ থেকে এই টপিক

Linear Combination, linear dependence & Independence

*) Vector $\underline{u}$ is a linear combination of vectors $\underline{v}_1, \underline{v}_2, \underline{v}_n \in V$ for any scalars $\alpha_1, \alpha_2, \ldots, \alpha_n$ such that

$\underline{u} = \alpha_1 \underline{v}_1 + \alpha_2 \underline{v}_2 + \ldots + \alpha_n \underline{v}_n = 0$

has only the trivial sol^n $\alpha_1 = 0, \alpha_2 = 0 \ldots, \alpha_n = 0$. If there are also nontrivial sol^n, then S is called linerly dependent.

1. At $\underline{v} = (5, 9, 5)$ and $\underline{v}_1 = (2, 1, 4)$, $\underline{v}_2 = (1, -1, 3)$, $\underline{v}_3 = (3, 2, 5)$.
Show that $\underline{v}$ can be expressed as a linear combination of vectors $\underline{v}_1$, $\underline{v}_2$ and $\underline{v}_3$.

Sol^n:-

$\underline{v}$ can be expressed as a linear combination of the vectors $\underline{v}_1$, $\underline{v}_2$ and $\underline{v}_3$ if for any scalar α_1, α_2 and α_3 such that

$$\underline{v} = \alpha_1 \underline{v}_1 + \alpha_2 \underline{v}_2 + \alpha_3 \underline{v}_3$$

$$\Rightarrow (5, 9, 5) = \alpha_1 (2, 1, 4) + \alpha_2 (1, -1, 3) + \alpha_3 (3, 2, 5)$$

$$\Rightarrow (5, 9, 5) = (2\alpha_1, \alpha_1, 4\alpha_1) + (\alpha_2, -\alpha_2, 3\alpha_2) + (3\alpha_3, 2\alpha_3, 5\alpha_3)$$

$$\Rightarrow (5, 9, 5) = (2\alpha_1 + \alpha_2 + 3\alpha_3, \alpha_1 - \alpha_2 + 2\alpha_3, 4\alpha_1 + 3\alpha_2 + 5\alpha_3)$$

ডিটেইলস এসাইনমেন্ট H.W

Equating the corresponding components, we get the following syste of linear equations,

$$\left.\begin{aligned} 2\alpha_1 + \alpha_2 + 3\alpha_3 &= 5 \\ \alpha_1 - \alpha_2 + 2\alpha_3 &= 9 \\ 4\alpha_1 + 3\alpha_2 + 5\alpha_3 &= 5 \end{aligned}\right\} \cdots\cdots (1)$$

$$\sim \begin{cases} 2\alpha_1 + \alpha_2 + 3\alpha_3 = 5 \\ -3\alpha_2 + \alpha_3 = 13 \\ 7\alpha_2 - 3\alpha_3 = -31 \end{cases} \quad \begin{bmatrix} L_2' = 2L_2 - L_1 \\ L_3' = L_3 - 4L_2 \end{bmatrix}$$

$$\sim \begin{cases} 2\alpha_1 + \alpha_2 + 3\alpha_3 = 5 \\ -3\alpha_2 + \alpha_3 = 13 \\ -2\alpha_3 = -12 \end{cases} \quad \left[L_3' = 3L_3 + 7L_2\right]$$

Using back substitution method from 3rd equation, we get $\alpha_3 = 1$

Substituting $\alpha_3 = 1$ in 2nd equdtion, we get

$$-3\alpha_2 + 1 = 13$$

$$\Rightarrow \alpha_2 = -4$$

From I st equation

$$2\alpha_1 - 4 + 3 = 5$$

$$\Rightarrow \alpha_1 = 3$$

Thus the soln of the system is

$$\alpha_1 = 3, \quad \alpha_2 = -4, \quad \alpha_3 = 1$$

Therefore $\underline{V} = 3\underline{v_1} - 4\underline{v_2} + \underline{v_3}$

Hence $\underline{v}$ can be expressed as a linear combination of $\underline{v_1}$, $\underline{v_2}$ and $\underline{v_3}$

{ এটা solution না এলাও
Hence $\underline{v}$ can not be expressed as a linear combination of $\underline{v_1}$, $\underline{v_2}$ and $\underline{v_3}$ (Final প্রশ্নের উত্তর)

25 Oct

Eigen value and Eigen Vector

1) A scalar λ is called an eigenvalue of a matrix A if there is a non-trivial solution $\underline{x}$ of $A\underline{x} = \lambda\underline{x}$ and $\underline{x}$ is called the eigen vector

2) $(\lambda I - A)$ or $(A - \lambda I)$ is called the characteristic matrix of A

3) $|\lambda I - A|$ or $|A - \lambda I|$ is called the characteristic polynomial of A

4) $|\lambda I - A| = 0$ or $|A - \lambda I| = 0$ is called the characteristic equation of A

Ex 1:
Find the characteristic matrix of

$$A = \begin{pmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{pmatrix}$$

Soln:

$$A - \lambda I$$

$$= \begin{pmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{pmatrix} - \begin{pmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{pmatrix}$$

$$= \begin{pmatrix} 2-\lambda & 2 & 1 \\ 1 & 3-\lambda & 1 \\ 1 & 2 & 2-\lambda \end{pmatrix}$$

is the characteristic matrix.

Ex 2:
Find the characteristic ~~matrix~~ polynomial of

Soln:

$$\begin{vmatrix} 2-\lambda & 2 & 1 \\ 1 & 3-\lambda & 1 \\ 1 & 2 & 2-\lambda \end{vmatrix}$$

$$= (2-\lambda)(6-5\lambda+\lambda^2-2) - 2(2-\lambda+1) + 1(2-3+\lambda)$$

$$= (2-\lambda)(\lambda^2-5\lambda+4) - 2(3-\lambda) + 1(\lambda-1)$$

$$= -\lambda^3+7\lambda^2-11\lambda+5\lambda^0$$

Cubic polynomial

Ex 3:-

Find the characteristic roots or Eigen values

Soln:-

Ex 2- এর solution সম্পর্ক নিম্নরূপ হবে —

$\lambda^3 + 7\lambda^2 + 11\lambda - 5 = 0$

$\Rightarrow \lambda^3 - \lambda^2 - 6\lambda^2 + 6\lambda + 5\lambda - 5 = 0$

$\Rightarrow \lambda^2(\lambda-1) - 6\lambda(\lambda-1) + 5(\lambda-1) = 0$

$\Rightarrow (\lambda-1)(\lambda^2 - 6\lambda + 5)$

$\Rightarrow (\lambda-1)(\lambda^2 - 5\lambda - \lambda + 5) = 0$

$\Rightarrow (\lambda-1)[\lambda(\lambda-5) - 1(\lambda-5)] = 0$

$\Rightarrow (\lambda-1)(\lambda-5)(\lambda-1) = 0$

$\lambda - 1 = 0$, $\lambda - 5 = 0$, $\lambda - 1 = 0$

$\Rightarrow \lambda = 1$, $\Rightarrow \lambda = 5$, $= \lambda = 1$

$\therefore \lambda = 1, 1, 5$

Repeated value থাকলেও লিখতে হবে

CT-6 প্রশ্নের

$\begin{matrix} 1 & 0 & -1 \\ 1 & 2 & -1 \\ 2 & 2 & 3 \end{matrix}$ 129

C.W Find the characteristic roots or Eigen values of $\begin{pmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{pmatrix}$

$\Rightarrow A - \lambda I$

$= \begin{pmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

$= \begin{pmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{pmatrix} - \begin{pmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{pmatrix}$

$= \begin{pmatrix} 6-\lambda & -2 & 2 \\ -2 & 3-\lambda & -1 \\ 2 & -1 & 3-\lambda \end{pmatrix}$

$|A - \lambda I| = \begin{vmatrix} 6-\lambda & -2 & 2 \\ -2 & 3-\lambda & -1 \\ 2 & -1 & 3-\lambda \end{vmatrix}$

Calculation Problem

$= (6-\lambda)\{(3-\lambda)^2 - 1\} + 2[-2(3-\lambda) + 1(2)] + 2[2 - 2(3-\lambda)]$

$= (6-\lambda)(9 - 6\lambda + \lambda^2 - 1) + 2(-6 - 2\lambda + 2) + 2(2 - 6 + 2\lambda)$

$= (6-\lambda)(8 - 6\lambda + \lambda^2) - 12 - 4\lambda + 4 + 4 - 12 + 4\lambda$

$= 48 - 36\lambda + 6\lambda^2 - 8\lambda + 6\lambda^2 - \lambda^3 - 12 - 4\lambda + 8 - 12 + 4\lambda$

$= 32 - \ldots\ldots\lambda + 12\lambda^2 - \lambda^3$

$= \lambda^3 - 12\lambda^2 + 36\lambda - 32$

Characteristic equation is $[A - \lambda I] = 0$

$\lambda^3 - 12\lambda^2 + 36\lambda - 32 = 0$

$\Rightarrow \lambda^3 - 2\lambda - 10\lambda^2 + 20\lambda + 16\lambda - 32 = 0$

$\Rightarrow \lambda^2(\lambda - 2) - 10\lambda(\lambda - 2) + 16(\lambda - 2) = 0$

$\Rightarrow (\lambda-2)(\lambda^2-10\lambda+16)=0$

$\Rightarrow (\lambda-2)(\lambda^2-8\lambda-2\lambda+16)=0$

$\Rightarrow (\lambda-2)\{\lambda(\lambda-8)-2(\lambda-8)\}=0$

$\Rightarrow (\lambda-2)(\lambda-8)(\lambda-2)=0$

$\therefore \lambda = 2, 2, 8$

H.W Find the charateristic roots or eigen values of $\begin{pmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{pmatrix}$

Sol^n:

$A-\lambda I$

$= \begin{pmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

$= \begin{pmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{pmatrix} - \begin{pmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{pmatrix}$

$= \begin{pmatrix} 1-\lambda & 0 & -1 \\ 1 & 2-\lambda & 1 \\ 2 & 2 & 3-\lambda \end{pmatrix}$

$|A-\lambda I| = \begin{vmatrix} 1-\lambda & 0 & -1 \\ 1 & 2-\lambda & 1 \\ 2 & 2 & 3-\lambda \end{vmatrix}$

$= (1-\lambda)\{(2-\lambda)(3-\lambda)-2\} - 1\{2-2(2-\lambda)\}$

$= (1-\lambda)\{6-2\lambda-3\lambda+\lambda^2-2\} - 1(2-4+2\lambda)$

$= (1-\lambda)(4-5\lambda+\lambda^2)+2-2\lambda$

$= 4-5\lambda+\lambda^2-4\lambda+5\lambda^2-\lambda^3+2-2\lambda$

$= -\lambda^3 + 6\lambda^2 - 11\lambda + 6$

characteristic equation is $[A - \lambda I] = 0$

$-\lambda^3 + 6\lambda^2 - 9\lambda + 4 = 0$

$\Rightarrow -\lambda^3 + \lambda^2 + 5\lambda^2 - 5\lambda - 4\lambda + 4 = 0$

$\Rightarrow -\lambda^2(\lambda - 1) + 5\lambda(\lambda - 1) - 4(\lambda - 1) = 0$

$\Rightarrow (\lambda - 1)(-\lambda^2 + 5\lambda - 4) = 0$

$\Rightarrow (\lambda - 1)(-\lambda^2 + \lambda + 4\lambda - 4) = 0$

$\Rightarrow (\lambda - 1)\{-\lambda(\lambda - 1) + 4(\lambda - 1)\} = 0$

$\Rightarrow (\lambda - 1)(\lambda - 1)(-\lambda + 4) = 0$

$-\lambda^3 + 6\lambda^2 - 11\lambda + 6 = 0$

$\Rightarrow -\lambda^3 + \lambda^2 + 5\lambda^2 - 5\lambda - 6\lambda + 6 = 0$

$\Rightarrow -\lambda^2(\lambda - 1) + 5\lambda(\lambda - 1) - 6(\lambda - 1) = 0$

$\Rightarrow (\lambda - 1)(-\lambda^2 + 5\lambda - 6) = 0$

$\Rightarrow (\lambda - 1)(-\lambda^2 + 3\lambda + 2\lambda - 6) = 0$

$\Rightarrow (\lambda - 1)\{-\lambda(\lambda + 3) - 2(\lambda - 3)\} = 0$

$\Rightarrow (\lambda - 1)(-\lambda - 2)(\lambda - 3) = 0.$

$\therefore \lambda = 1, 2, 3$

27/10/2026

iterative method. pdf

Final ও 10 নম্বরে আসতে পারে Jacobi Gauss

~~The Jacobi method:~~

Ex:- Apply the jacobi method to solve

$$5x_1 - 2x_2 + 3x_3 = -1$$
$$-3x_1 + 9x_2 + x_3 = 2$$
$$2x_1 - x_2 - 7x_3 = 3$$

[An iterative process involves systematically repeating a sequence of or formula to achieve the desired result. Through trial and error, you may get closer to a solution.]

Continue iterations until two successive approximations are identical (rounded to 3 significant digits)

$$\begin{cases} 5x_1 = -1 + 2x_2 - 3x_3 \\ 9x_2 = 2 + 3x_1 - x_3 \\ -7x_3 = 3 - 2x_1 + x_2 \\ (7x_3 = -3 + 2x_1 - x_2) \end{cases}$$

$$\therefore x_1 = \frac{1}{5}(-1 + 2x_2 - 3x_3)$$
$$x_2 = \frac{1}{9}(2 + 3x_1 - x_3)$$
$$x_3 = \frac{1}{7}(-3 + 2x_1 - x_2) \quad \text{---} (*)$$

Consider $x^{(0)} = (0, 0, 0)$

$$x_1^{(1)} = \frac{1}{5}(-1 + 0 - 0) = -\frac{1}{5} = -0.200$$
$$x_2^{(1)} = \frac{1}{9}(2 + 0 - 0) = \frac{2}{9} = 0.222$$
$$x_3^{(1)} = \frac{1}{7}(-3 + 0 - 0) = -0.428$$

$x^{(1)} = (x_1^{(1)}, x_2^{(1)}, x_3^{(1)}) = (-0.200, 0.222, -0.428)$

Now we start the 2nd iteration,

$$x_1^{(2)} = \frac{1}{5}(-1 + 2(0.222) - 3(-0.428))$$

$$= \frac{1}{5}(-1 + 0.444 + 1.284) = \frac{1}{5}(0.728) = 0.1456$$

$$x_2^{(2)} = \frac{1}{9}(2 + 3(-0.200) + 0.428)$$

$$= \frac{1}{9}(2 - 0.6 + 0.428) = \frac{1}{9}(1.828) = 0.203$$

$$x_3^{(2)} = \frac{1}{7}(-3 + 2(-0.200) - 0.222) = \frac{1}{7}(-3 - 0.4 - 0.222)$$

$$= \frac{1}{7}(-3.622)$$

$$= -0.517$$

For 3rd iteration, we will use the values of 2nd iteration

$$x_1^{(3)} = \frac{1}{5}(-1 + 2(0.203) - 3(-0.517))$$

$$= \frac{1}{5}(-1 + 0.406 + 1.551) = \frac{1}{5}(0.957) = 0.1914$$

$$x_2^{(3)} = \frac{1}{9}(2 + 3(0.1456) - (-0.517))$$

$$= \frac{1}{9}(2 + 0.4368 + 0.517) = \frac{1}{9}(2.9538) = 0.3282$$

$$x_3^{(3)} = \frac{1}{7}(-3 + 2(0.1456) - 0.203)$$

$$= \frac{1}{7}(-3 + 0.2912 - 0.203) = -\frac{1}{7}(0.2912 - 3.203)$$

$$= -\frac{0.2912}{7} = -0.416$$

$\therefore x^{(3)} = (0.1914, 0.3282, -0.416)$

For 4th iteration, we will use the values of 3rd iteration

$$x_1^{(4)} = \frac{1}{5}(-1+2(0.3282)-3(-0.416))$$

$$= \frac{1}{5}(-1+0.6564+1.248) = \frac{1}{5}(0.9044) = 0.1808$$

$$x_2^{(4)} = \frac{1}{9}(2+3(0.1914)-(-0.416))$$

$$= \frac{1}{9}(2+0.5742+0.416) = \frac{1}{9}(2.9902) = 0.332$$

$$x_3^{(4)} = \frac{1}{7}(-3+2(0.1914)-0.3282)$$

$$= \frac{1}{7}(-3+0.3828-0.3282) = \frac{1}{7}(0.3828-0.3282)$$

$$= -0.421$$

$$x^4 = (0.1808, 0.3328, -0.421)$$

For 5th iteration,

$$x_1^{(5)} = \frac{1}{5}(-1+2(0.332)-3(-0.421)$$

$$= \frac{1}{5}(-1+0.664+1.263) = \frac{1}{5}(0.927) = 0.1854$$

$$x_2^{(5)} = \frac{1}{9}(2+3(0.1808)+0.421)$$

$$= \frac{1}{9}(2.421+0.5424) = 0.329$$

$$x_3^{(5)} = \frac{1}{7}(-3+2(0.1808)-0.332)$$

$$= \frac{1}{7}(-3.332+0.329, -0.424)$$

$$x^{(5)} = (0.1854\ 0.329, -0.424)$$

For 6th iteration,

$$x_1^{(6)} = \frac{1}{5}(-1 + 2(0.329) - 3(-0.424))$$

$$= \frac{1}{5}(-1 + 0.658 + 1.272)$$

$$= \frac{0.93}{5} = 0.186$$

$$x_2^{(6)} = \frac{1}{9}(2 + 3(0.1854) + 0.424)$$

$$= \frac{1}{9}(2 + 0.5562 + 0.424) = 0.331$$

$$x_3^{(6)} = \frac{1}{7}(-3 + 2(0.1854) - 0.329)$$

$$= \frac{1}{7}(-3 + 0.3708 - 0.329) = \frac{-2.9582}{7} = -0.422$$

$$x^{(6)} = (0.186, 0.331, -0.422)$$

In 5th iteration and 6th iteration, two successive approximation are identical.

Ex:- Apply Gauss-seidal method to solve,

$$5x_1 - 2x_2 + 3x_3 = -1$$
$$-3x_1 + 9x_2 + x_3 = 2$$
$$2x_1 - x_2 - 7x_3 = 3$$

$$\therefore x_1 = \frac{1}{5}(-1 + 2x_2 - 3x_3)$$
$$x_2 = \frac{1}{9}(2 + 3x_1 - x_3)$$
$$x_3 = \frac{1}{7}(-3 + 2x_1 - x_2)$$

Consider the initial approximation $x^{(0)} = (0, 0, 0)$

$$x_1^{(1)} = \frac{1}{5}(-1 + 0 - 0) = -0.2$$

$$x_2^{(1)} = \frac{1}{9}(2 + 3(-0.2) - 0) = \frac{1}{9}(2 - 0.6) = 0.155$$

$$x_3^{(1)} = \frac{1}{7}(-3 + 2(-0.2) - 0.155)$$

$$= \frac{1}{7}(-3 - 0.4 - 0.155) = \frac{-3.555}{7} = -0.507$$

$$x^{(1)} = (-0.2, 0.155, -0.507)$$

For 2nd iteration, we will consider values of $x^{(1)}$

$$x_1^{(2)} = \frac{1}{5}(-1 + 2(0.155) - 3(-0.507))$$

$$= \frac{1}{5}(-1 + 0.31 + 1.521) = \frac{0.831}{5} = 0.166$$

$$x_2^{(2)} = \frac{1}{9}(2 + 3(0.166) + 0.507)$$

$$= \frac{1}{9}(2 + 0.498 + 0.507) = \frac{3.0056}{9} = 0.334$$

$$x_3^{(2)} = \frac{1}{7}(-3 + 2 \times 0.166 - 0.334) = \frac{1}{7}(-3 + 0.332 - 0.334)$$

$$= \frac{1}{7}(-3.002) = -0.428$$

$$x^{(2)} = (0.166, 0.334, -0.428)$$

We start 3rd iteration,

$x_1^{(3)} = \frac{1}{5}(-1+2(0.334) - 3(-0.428))$

$= \frac{1}{5}(-1+0.668+1.284) = \frac{0.952}{5} = 0.1904$

$x_2^{(3)} = \frac{1}{9}(2+3(0.1904)+0.428)$

$= \frac{1}{9}(2+0.5712+0.428) = 0.333$

$x_3^{(3)} = \frac{1}{7}(-3+2(0.1904)-0.333)$

$= \frac{1}{7}(-3+0.3808-0.333) = \frac{1}{7}(0.3808-3.333)$

$x^{(3)} = (0.1904, 0.333, -0.421) = \frac{-2.9522}{7} = -0.421$

We start 4th iteration,

$x_1^{(4)} = \frac{1}{5}(-1+2(0.333)-3(-0.421)$

$= \frac{1}{5}(-1+0.666+1.263) = 0.1858$

$x_2^{(4)} = \frac{1}{9}(2+3(0.1858)+0.421)$

$= \frac{1}{9}(2+0.5574+0.421) = 0.3309$

$x_3^{(4)} = \frac{1}{7}(-3+2(0.1858)-0.3309)$

$= \frac{1}{7}(-3+0.3716-0.3309)$

$= \frac{1}{7}(0.3716-3.3309)$

$= -0.422$

$x^{(1)} = (0.1858, 0.33091, -0.422)$

We start 5th iteration,

$x_1^{(5)} = \frac{1}{5}(-1+2(0.3309)-3(-0.422))$

$= \frac{1}{5}(-1+0.6618-1.266) = 0.1855$

$x_2^{(5)} = \frac{1}{9}(2+3(0.1855)+0.422)$

$= \frac{1}{9}(2+0.5566+0.422) = 0.3309$

$x_3^{(5)} = \frac{1}{7}(-3+2(0.1855)-0.3309)$

$= \frac{1}{7}(-3+0.371-0.3309)$

$= \frac{1}{7}(0.371-3.3309)$

$= -0.422$

$x^{(5)} = (0.1855, 0.3309, -0.422)$

H.W

* $26x_1 + 2x_2 + 2x_3 = 12.6$
  $3x_1 + 27x_2 + x_3 = -14.3$
  $2x_1 + 3x_2 + 17x_3 = 6$

Jacobi method and Gauss-seidal meth

* $45x_1 + 2x_2 + 3x_3 = 58$
  $-3x_1 + 22x_2 + 2x_3 = 47$
  $5x_1 + x_2 + 20x_3 = 67$

~~Solution~~ Answer slide 10 ও আছে

H.W assignment